School of Information Technology and Data Science

Occupational Certificate: Software Engineer (NQF5) Software Engineering (SOEN)	
Curriculum Code	251201001
SAQA Qualification ID	119458
Intake / year	2-2025
Assessment Name	Summative Assessment
Module NQF Level	6
Credits	20
Open date and time	29 August 2025 • ***Section A: 17:15*** • ***Section B: 18:45***
Close date and time	29 August 2025 • ***Section A: 18:45*** • ***Section B: 20:15***
Assessment Total Marks	Section A = 70 (20% of total mark percentage) Section B = 60 (30% of total mark percentage)
Pass percentage	65%

Lecturer Information	
Lecturer	Preciouse Nkabinde
Lecturer E-mail	PreciouseN@aie.ac

IMPORTANT NOTICE

Before submitting your assessment, please ensure that it is signed and that all your personal details are accurately filled in the designated space below (Section in red). Work will not be accepted unless the required sections below are fully completed prior to submission, resulting in a 0% grade for this assessment.

The signature must be your valid handwritten signature and NOT your name and surname typed in the space provided.

Student Number:	360743		
Student name and surname:	**Kgutlane Ralijeje**	**Date:**	**27/03/2026**
Student cell phone number:	**0837935476**		
Student Signature:			
Lecturer name and surname:		**Date:**	
Lecturer Signature:			
Assessor name and surname:		**Date:**	
Assessor Signature:			
Moderator name and surname:		**Date:**	
Moderator Signature:			

Instructions to Candidates
1. Read each question carefully. 2. You must answer Section B below the space provided named "STUDENTANSWER". Once completed, the answer sheet, with your answers, must be converted to PDF and uploaded to the student portal. Make sure that all additional information is uploaded in a single Zipped folder. 3. QCTO students who achieve between 55-64% may have a further submission opportunity, subject to the departmental rules. 4. NO LATE submissions will be accepted. (System will automatically close when due date and time is reached). If you encounter any difficulty with the student portal, immediately take a screenshot, log a ticket (attention support), and attach your assessment to the ticket. 5. You must follow the departmental formatting guidelines. 6. You must write your name, surname, and student number on your script. A failure to do so will result in your script not being marked. 7. By submitting your answer sheet for marking, you declare the following: a. You are familiar with the AIE's policies on academic dishonesty and plagiarism; b. Your submitted is your own original work. Where you have made use of someone else's work, you have duly referenced such work making using of the APA style of referencing; c. You have not allowed, and will not allow any other student to copy your work; and d. You did not copy the work of another student. 8. Students that are dishonest will be subjected to disciplinary procedures.

Assessment Objective/s:

The Summative assessment (Section A and B) will cover the following assessment topics:

No	KM/KT number	KT Description
1.	KM01-KT0101	Overview/ introduction of system analysis and design
2.	KM01-KT0102	System Planning
3.	KM01-KT0103	System Analysis
4.	KM01-KT0104	System Design
5.	KM01-KT0105	System Documentation
6.	KM01-KT0201	Software and System Testing Methodologies
7.	KM01-KT0202	Software and System Testing Techniques
8.	KM01-KT0203	Software and System Testing Documentation Procedures
9.	KM01-KT0301	System integration methodologies and processes
10	KM01-KT0302	System Deployment Methodologies and processes
11	KM01-KT0303	System Deployment and integration documentation
12	KM01-KT0401	System Maintenance Procedures
13	KM01-KT0402	System Management Tools

Scope:

The primary aim of this module is to equip you with a comprehensive understanding of software engineering principles. By exploring systems analysis and design, software and system testing, system deployment and integration, and system management, you will gain

the knowledge and skills needed to navigate the complexities of developing and maintaining software systems effectively.

Technical Aspects:

Please verify that all pages are included when downloading this assessment by checking the page numbers at the bottom of the document.

Ensure the following font and size is used in this assessment

a. Font: Arial
b. Size: 12
c. Font colour: Black

Save and upload the report as a .PDF **(No backgrounds)** with the following naming convention:

a. StudentName_ StudentSurname _Student no_ModuleCode_FA1

Mark allocation for report

See Mark allocation sheet below

Question 1 (5)

KM No	*KT No*	*KT Description*
01	*0101*	*Overview/ introduction of system analysis and design*

a) Identify five (5) benefits of system analysis and design [5]

1. Enabling comprehension of complex structures.
2. Allowing for the better management of any business changes.
3. Aligning the organization with it's environment and strategic priorities.
4. Reducing IT issues and also reducing the workload of IT employees.
5. Improving the overall quality of a system.

Question 2 (7)

KM No	*KT No*	*KT Description*

01	0102	System Planning

a) Illustrate with the aid of a sketch the SDLC. [7]

Phase 1: Planning - This is project identification and aligning resources with strategic goals.

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Phase 2: Analysis - This is the gathering of data, identifying issues, and making determination of user requirements.

↓

Phase 3: Design - Creating the blueprint (core components, architecture, modules, data elements, and interfaces)

↓

Phase 4: Implementation - The building and coding of the software components

↓

Phase 5: Testing - Verifying and validating that the system meets all design and requirement specifications.

↓

Phase 6: Deployment - The roll-out and transition of the system to the live environment and end users.

↓

Phase 7: Maintenance - Ongoing system support, fault repair, and system updates.

Question 3 (16)

KM No	KT No	KT Description
01	0103	System Analysis

a) Elaborate why systems analysis is important. [16]

Systems analysis can be described as the structured or detailed process of studying an existing or proposed system to analyze, model, and recommend logical alternatives for improvements. This serves as the bridge between identifying a business problem and designing a tailored technical solution.

1. Managing Development complexity

One of the most important reasons for systems analysis is to alleviate complexity. By utilizing a structured approach stakeholders can decompose a large, complicated system into smaller, manageable, and measurable components.

2. Risk Mitigation through Feasibility analysis

Systems analysis entails a rigorous feasibility analysis to ensure the project is worth the investment, It evaluates the project across four key criteria:

1. Technical
2. Economic
3. Operational
4. Schedule

3. Requirement Gathering

Systems analysis ensures that the “what” of the system is clearly defined before the “how”(design). It often entails gathering both functional and non-functional requirements directly from stakeholders.

4. Alignment with strategic goals

Systems analysis makes sure that any proposed software solution aligns with the organizations strategic objectives.

Question 4 (16)

KM No	*KT No*	*KT Description*
01	*0105*	*Viruses*

a) According to Heidi Adkisson, Principal UX Designer & Partner at Blink UX, while there are many different design docs variants. [10]

Identify and discuss five (5) of the more task-specific types.

1. **User stories**: This defines functionality from the users perspective to guide the design decisions.

2. **Screen Flow Diagrams:** This is the visualization of user navigation across system interfaces.

3. **Use Cases**: Present comprehensive narritives about system usage scenarios.

4. **Page-Level Documentation**: This elaborates the purpose and function of specific UI pages.

5. **Scenario Narratives:** Provides contextual walkthroughs of tasks within the system.

b) Identify six (6) purposes of system documentation. [6]

1. Provide reference material to support training and onboarding of users and operators
2. Function as a development checkpoint for tracking progress or environments.
3. Simplify the migration of systems between platforms or enviroments.
4. Facilitates easier modifications, upgrades and system integrations.
5. Improves communications between developers, users, and business stakeholders.
6. Allows advance planning by detailing expected system behaviors and deliverables.

Question 5 (8)

KM No	KT No	KT Description
01	0201	Software and System Testing Methodologies

a) What is your understanding of what white box testing is? [5]

White box testing is a type of software testing method where the internal structure, design, and implementation of the item or system being tested are known to the tester.

Furthermore, they have the following key aspects:

1. Internal visibility where the focus is on the internal logic and flow of the software.

2. Code Coverage where the primary goal is to make sure that all internal operations are performed according to the specifications

3. Security and hidden errors where it is highly effective in uncovering hidden vulnerabilities, logical errors and dead code.

b) Identify three (3) main techniques included in white box testing. [3]

1. Statement coverage

2. Branch coverage

3. Path coverage

Question 6 (8)

KM No	KT No	KT Description
01	0203	Software and System Testing Documentation Procedures

a) Identify eight (8) benefits of using documentation. [8]

1. **I**t clarifies the quality of methods and objectives.
2. It ensires internal coordimation when a customer uses software application.
3. It ensures clarity about the stability of tasks and performance.
4. It provides feedback on preventative tasks.
5. It provides feedback for your planning cycle.
6. It creates objective evidence fir the performance of the quality management system.
7. It is a time-saving process because you can easily refer to the text document.
8. It is consistent because you can test the same value.

Mark allocation for student		
Question	**Maximum Mark**	**Learner mark**
Question 1.a	5	
Question 2.a	7	
Question 3.a	16	
Question 4.a	10	
Question 4.b	6	
Question 5.a	5	
Question 5.b	3	
Question 6.a	8	

Total:	60	

STUDENT ASSESSMENT FEEDBACK

Qualification Name:	**Occupational Certificate: Software Engineer (NQF5)**
Qualification SAQA Number:	**119458**
Subject Name:	**Software Engineering**
Subject Code:	**SOEN**
Assessment Name:	**Summative Assessment**
Assessment Code:	**SOEN_SA**

Assessment Decision	**Competent**		**Not yet competent**	

Feedback report	1st Attempt		2nd Attempt	
	C	NYC	C	NYC
Unit standard Number(s)				
KM01				
KT0101				
KT0102				
KT0103				
KT0104				
KT0105				
KT0202				
KT0203				
KT0301				
KT0302				
KT0303				
KT0401				
KT0402				

General feedback to learner **Supply comprehensive feedback why learner is found NYC**